# CHANDU PRASAD BHAIRAPU



## AI Engineer

bhairapuchanduprasad@gmail.com (+49) 15560194877 Frankfurt • EU Blue Card linkedin github

## ABOUT ME

Master's graduate in Artificial Intelligence with 3 years of professional experience in AI engineering and software development. Specialized in data-driven solutions from problem definition to evaluation, combining machine learning and forecasting with generative AI and RAG. Most recently sole developer of an agentic knowledge-extraction system for the German energy sector, backed by data science work on industrial process data.

## EXPERIENCE

### Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

**Kniit GmbH**

November 2025 – present | Muldestausee, Germany

- Sole developer owning the full knowledge-extraction pipeline end-to-end – data ingestion, chunking, prompt engineering, and indexing (agentic, deterministic, and monolithic strategies) – reporting directly to the CTO and collaborating with SAP domain experts across the German energy sector.
- Designed and optimized multi-agent extraction algorithms (LangGraph: entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for knowledge extraction from complex regulatory and technical documents, driving precision/recall from 33%/45% to 93%/90% via an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) extraction pipeline for a second, high-volume source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph for millisecond-latency retrieval, integrated directly into Claude Code for agentic development workflows.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) deployed on Azure, enabling domain experts to validate and approve knowledge-graph changes.
- Delivered as an AI-augmented engineer – leveraging Claude Code, custom skills, and agentic coding workflows for rapid, high-quality development grounded in clean code principles.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, RAG, GraphRAG, MCP, Neo4j, MongoDB, Postgres, FastAPI, Docker, Kubernetes, Azure, React**

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### GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

**Covestro Deutschland AG**

December 2024 – April 2025 | Leverkusen, Germany

- Worked with business stakeholders to understand their decision-making workflows, advised where agentic AI could support them and translated their requirements into the solution approach.
- Designed and deployed an agentic AI application using LangGraph to optimize decision-making workflows for vector database queries, general inference, and summarization tasks.
- Developed a React-based front-end and a Python backend with FastAPI, leveraging Postman for API testing.
- Built and containerized the entire stack with Docker and deployed it to AWS using CDK, adhering to clean code practices. Version-controlled via GitLab, with CI/CD integration.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD, React**

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### GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

**Covestro Deutschland AG**

May 2024 – November 2024 | Leverkusen, Germany

- Automated RAG's Generation and Retrieval evaluation with RAGAS and PytrecEval, comparing model answers to chemist-provided ground truth.
- Built AWS cloud infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- Benchmarked multilingual embedding models on over 1M data chunks of Covestro's internal chemical data.
- Designed RAG pipelines, focusing on chunking strategies, vector databases, indexing, and fine-tuning models for optimized retrieval.

- **Tech stack: Python, RAGAS, PytrecEval, LangChain, Hugging Face Transformers, PyTorch, Scikit-Learn, AWS, Azure AI Search**

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### Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

**Covestro Deutschland AG**

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost and LSTM models to predict prepolymer synthesis outcomes (viscosity, reaction progression) from time-series process data; benchmarked against empirical curve-fitting, the models improved early identification of batches likely to deviate from spec.
- Engineered lag features and rolling statistical windows from sensor time series for the tree models and fed raw sequences into the LSTM; data preparation and analysis with Pandas, NumPy and SciPy.
- Delivered a Streamlit dashboard where chemists simulate parameter changes and see predictions before running physical batches, turning model results into recommendations they could act on.
- **Tech stack: Python, Scikit-Learn, XGBoost, TensorFlow/Keras, Pandas, NumPy, SciPy, Matplotlib, Streamlit**

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### Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

**Cognizant Technology Solutions**

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet and LSTM forecasting models for EV load time series; the LSTM reached 96% test accuracy with strong performance on unseen data.
- **Tech stack: Python, Scikit-Learn, TensorFlow**

## PROJECTS

**Project: Analysis of Life Expectancy with Regression Models | Python | Scikit-Learn**
- Statistical analysis of life expectancy data with regression models, visualization and feature selection to identify the main influencing factors.

**Project: Clinical Info Assistant | Python | FastAPI | LLMs | React**
- Web application that answers questions about medicines from drug label texts, with cited sources, a doctor handoff for patient questions, usage events and a funnel view to study user behavior, and a labelled evaluation set run in CI.

## TECHNICAL SKILLSET

- **Data Science & Statistics:** Python, SQL, Pandas, NumPy, SciPy, Matplotlib, feature engineering, regression, model evaluation, time-series forecasting (ARIMA, SARIMA, Prophet, LSTM)
- **Machine Learning:** Scikit-Learn, XGBoost, Random Forest, PyTorch, TensorFlow/Keras, Hugging Face Transformers
- **Generative AI:** Large Language Models, RAG, GraphRAG, AI agents (LangGraph, LangChain), prompt engineering, LiteLLM, MCP, RAGAS, QLoRA fine-tuning
- **Engineering & Cloud:** FastAPI, APIs, Streamlit, Docker, Kubernetes, Azure, AWS, Git, GitLab CI/CD, Postgres, MongoDB, Neo4j, vector databases

## EDUCATION

**Master of Science in Artificial Intelligence** — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

**Bachelor of Technology in Mechanical Engineering** — August 2017 – July 2021
Minor in Computer Science & Engineering
G Pulla Reddy Engineering College, Kurnool, India

## PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT

## LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)